

UESTC 2004 – Embedded Processors (Main Assignment Marking Scheme)

Overall Mark Distribution

Component	Weight
Technical Report	70%
Code + Video Demonstration	30%

Table 1: Technical Report Marking Rubric (70% of total)

Section	Weight (%)	High Quality (70%+)	Medium Quality (50-69%)	Low Quality (<50%)
Introduction	5	Clear project overview, objectives, scope. Includes relevant background. Motivation is convincing and clearly connected to the project.	Overview present but vague; weak or missing discussion of literature; motivation unclear.	No clear objectives; no references; no motivation; off-topic or missing.
Methodology	30	Requirements and specifications clearly listed. Both block diagram & flowcharts included. Hardware/software design justified (e.g.,	Some diagrams but missing justification. Power-aware decisions mentioned but not justified. Testing described	No design details. No diagrams. No justification. No testing strategy.

Section	Weight (%)	High Quality (70%+)	Medium Quality (50-69%)	Low Quality (<50%)
		why specific hardware? Why thresholds?). Power-aware decisions explained (sampling rates, sleep modes, interrupt use). Modular implementation used and described. Testing strategy (unit, integration, white box, black box) with clear pass/fail criteria. If any experimental procedures were used, they are clearly described. Clear description of the individual task allocations.	but vague. No clear modular structure. Vague description of individual task allocations.	Power not mentioned. No discussion of individual task allocations.
Results & Discussion	20	Presents actual data (e.g., serial output screenshots, temperature logs, event flags). Discusses significance of results – did the system meet requirements? Explains any unexpected behaviour. Links back to design choices.	Some results but superficial. Discussion lacks depth – just describes what happened, not why. No link to requirements/design.	No data shown. No discussion. Just claims "it worked".

Section	Weight (%)	High Quality (70%+)	Medium Quality (50-69%)	Low Quality (<50%)
Conclusions & Lessons Learnt	10	Summarises key findings. Discusses limitations (e.g., power draw, sensor accuracy, memory). Suggests realistic future improvements. Reflects on teamwork/lessons.	Conclusions present but generic. Limitations vague or missing. Future work unrealistic.	No conclusion. No reflection. No lessons learnt/future improvements.
References	5	Citation Quality: Consistent, correct, relevant sources. Correct use of IEEE style. 5 to 10 literature references are cited in the introduction. Use of sources (Did the student actually use references to support and justify their engineering decisions, or did they just list them?): References supports technical discussion	Citation Quality: Minimal issues. Less than 5 references are discussed. Use of sources: Minimal usage Use a clear but not standard IEEE citation style	Citation Quality: Missing or poor. Use of sources: Not used

Table 2: Code + Video Demo Marking Rubric (30% of total)

Component	Weight (%)	High Quality(70%+)	Medium Quality(50-69%)	Low Quality (<50%)
Functionality	10	System works as specified: reads TMP102, reads ADXL345, detects temperature exceedance, detects motion/handling, logs events, indicates status (LEDs/buzzer/display/serial). Low-power operation evident (or convincingly explained).	Partial functionality – e.g., temperature works but motion faulty. Events not reliably detected. No low-power consideration.	Does not work. Sensors not reading. No event detection. Code does not compile or run.
Code Quality & Modularity	5	Meaningful variable names. Block comments at top, line comments where needed. Consistent indentation. Code broken into functions/classes (e.g., <code>readTemperature()</code> , <code>checkMotion()</code> , <code>logEvent()</code>). No "magic numbers". Avoids <code>delay()</code> where blocking hurts power.	Some good practice but inconsistent. Long <code>main()</code> with repeated code. Some unclear names. Missing comments. Some use of indentation but inconsistent.	Poor structure. No comments. Unreadable. Single huge loop. No indentation.

Component	Weight (%)	High Quality(70%+)	Medium Quality(50-69%)	Low Quality (<50%)
Video Demonstration	10	3-5 mins. All group members appear/voice with equal involvement. Shows full function/operation: power-on, sensor readings, triggering events (e.g., warming sensor with finger, shaking board), indicator response. Explains what is happening. Shows serial output if used.	Too short or too long. Shows only idle state. Poor audio or unclear explanation. Only partial functionality demonstrated. Unequal involvement of team members.	No video. Video doesn't show working system. No explanation. Not all members present.
Understanding (evident in video)	5	Student explains <i>why</i> thresholds chosen, <i>how</i> power saving works, <i>what</i> event means. Can answer (in video) how they tested. Shows debugging awareness.	Basic description of "this reads sensor" but no deeper reasoning.	Cannot explain code. Reads from script without understanding.